YEAR 11 PHYSICS, AQA GCSE, LESSON 6 OF 10

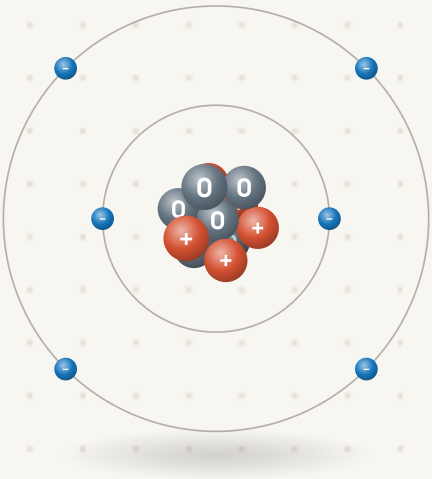
# Atoms & Nuclear Radiation.

## Lesson 06 Nuclear equations

Radium-226 decays and becomes radon, a completely different element. How can we predict exactly which nucleus it turns into?

IN THIS LESSON YOU WILL

- use the symbols for alpha, beta, gamma and neutron radiation
- balance a nuclear equation using mass numbers and atomic numbers
- work out the new nucleus after alpha or beta decay
- explain the effect of each decay on the mass and charge of the nucleus



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

# How this booklet works.

This is the booklet for **Lesson 6** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

**DO NOW**
Quick questions from earlier lessons. Remembering again is what makes it stick.

**LEARN IT**
Explanations, diagrams and worked examples. Read this again when you revise.

**TRY IT**
Practice that builds up: first with support, then on your own, then a challenge.

**LAB**
Record what you find in a simulation or a model experiment.

**EXAM CORNER**
AQA-style questions with a tip on how the marks are given.

**REVISE IT**
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

## The route through the unit (this lesson is highlighted)


01 Inside the atom

02 Isotopes and ions

03 From plum pudding to the nucleus

04 Energy levels and the nucleus

05 Radioactive decay

**06 Nuclear equations**
Symbols for each radiation, balancing mass and atomic numbers (slides 28–31)

07 Properties and uses of radiation

08 Half-life and random decay

09 Half-life calculations

10 Contamination and irradiation


Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

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LESSON

# 06 Nuclear equations.

AQA Physics 4.4.2.2; Combined Science 6.4.2.2

BIG QUESTION

**Radium-226 decays and becomes radon, a completely different element. How can we predict exactly which nucleus it turns into?**

BY THE END YOU CAN

- use the symbols for alpha, beta, gamma and neutron radiation
- balance a nuclear equation using mass numbers and atomic numbers
- work out the new nucleus after alpha or beta decay
- explain the effect of each decay on the mass and charge of the nucleus

nuclear equation | parent nucleus | daughter nucleus | mass number | atomic number | balance

SLIDES AND SIMULATIONS

**Teaching deck, slides 28–31**

Alpha, beta and gamma equations, and the 'What changes?' table.

panphy.app/gcsephy/decks/atoms-and-radiation/#28

**Nuclear Decay simulation**

panphy.app/simulations/nuclear_decay.html

RETRIEVE

## Do now

From earlier lessons. No notes!

1 What is an alpha particle made of?

2 What happens in the nucleus in beta decay?

3 What is the unit of activity?

EXPLAIN

# Balancing nuclear equations

## Symbols for nuclear radiation

| Emitted | Symbol | Mass number | Atomic number |
|---|---|---|---|
| Alpha particle | $^{4}_{2}\mathrm{He}$ | decreases by 4 | decreases by 2 |
| Beta particle | $^{0}_{-1}\mathrm{e}$ | no change | increases by 1 |
| Gamma ray | $\gamma$ | no change | no change |
| Neutron | $^{1}_{0}\mathrm{n}$ | decreases by 1 | no change |

An electron has almost no mass, so its top number is 0. Its charge is −1, so its bottom number is −1. When a neutron (charge 0) turns into a proton (charge +1), the beta particle carries away the −1: the charges still balance.

! In every nuclear equation, the **top numbers add up to the same total** on both sides, and so do the **bottom numbers**. If the atomic number changes, a **new element** forms.

## The three equations from the slides

$$^{241}_{95}\mathrm{Am} \rightarrow {}^{237}_{93}\mathrm{Np} + {}^{4}_{2}\mathrm{He}$$

Alpha. Top: 241 = 237 + 4; bottom: 95 = 93 + 2

$$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$$

Beta. Top: 14 = 14 + 0; bottom: 6 = 7 + (−1)

$$^{99\mathrm{m}}_{43}\mathrm{Tc} \rightarrow {}^{99}_{43}\mathrm{Tc} + \gamma$$

Gamma. The ‘m’ means the nucleus has extra energy; A and Z do not change

WORKED EXAMPLE

Radium-226, $^{226}_{88}Ra$, emits an alpha particle and becomes radon (Rn). Write the equation.

| | |
|---|---|
| TOP | 226 − 4 = 222 |
| BOTTOM | 88 − 2 = 86 |
| NEW NUCLEUS | $^{222}_{86}Rn$ |
| EQUATION | $^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + {}^{4}_{2}He$ |

YOUR TURN

Uranium-238, $^{238}_{92}U$, emits an alpha particle and becomes thorium (Th). Write the equation.

| | |
|---|---|
| TOP | |
| BOTTOM | |
| NEW NUCLEUS | |
| EQUATION | |

WORKED EXAMPLE

Strontium-90, $^{90}_{38}Sr$, emits a beta particle and becomes yttrium (Y). Write the equation.

| | |
|---|---|
| TOP | 90 − 0 = 90 |
| BOTTOM | 38 + 1 = 39 |
| NEW NUCLEUS | $^{90}_{39}Y$ |
| EQUATION | $^{90}_{38}Sr \rightarrow {}^{90}_{39}Y + {}^{0}_{-1}e$ |

YOUR TURN

Iodine-131, $^{131}_{53}I$, emits a beta particle and becomes xenon (Xe). Write the equation.

| | |
|---|---|
| TOP | |
| BOTTOM | |
| NEW NUCLEUS | |
| EQUATION | |

PRACTISE

## Try it

1 SUPPORTED Fill in the missing particle. Check that the top and bottom numbers balance.

$$^{210}_{84}Po \rightarrow {}^{206}_{82}Pb + \square$$

$$^{60}_{27}Co \rightarrow {}^{60}_{28}Ni + \square$$

**2** ON YOUR OWN Fill in the missing nucleus. The element symbols are given.

$$^{222}_{86}\text{Rn} \rightarrow \square + {}^{4}_{2}\text{He}$$

*Po = polonium*

$$^{137}_{55}\text{Cs} \rightarrow \square + {}^{0}_{-1}\text{e}$$

*Ba = barium*

$$^{5}_{2}\text{He} \rightarrow \square + {}^{1}_{0}\text{n}$$

*a helium-5 nucleus emits a neutron*

**3** Name the type of decay in each equation.

| Equation | Type of decay |
|---|---|
| $^{40}_{19}\text{K} \rightarrow {}^{40}_{20}\text{Ca} + {}^{0}_{-1}\text{e}$ | |
| $^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + {}^{4}_{2}\text{He}$ | |
| $^{99m}_{43}\text{Tc} \rightarrow {}^{99}_{43}\text{Tc} + \gamma$ | |

**4** Complete the table to show the effect of each decay on the nucleus.

| Radiation emitted | Effect on the mass of the nucleus | Effect on the charge of the nucleus |
|---|---|---|
| alpha | | |
| beta | | |
| gamma | | |

**5** Explain what is wrong with Leo's statement.

**Leo**: In beta decay the nucleus loses an electron, so the mass number goes down by 1.

6 CHALLENGE Uranium-238, $^{238}_{92}U$, decays by alpha emission. The new nucleus then decays by beta emission, and the next one by beta emission again. Find the final nucleus. What do you notice?

AQA-STYLE

## Exam corner

Write in full sentences. Show every step of a calculation.

**Get the marks:** Both numbers earn a mark. Check: top 212 = ? + 4, bottom 83 = ? + 2.

7 Bismuth-212, $^{212}_{83}Bi$, decays by alpha emission to form thallium (Tl). Complete the nuclear equation. **[2 marks]**

$$^{212}_{83}Bi \rightarrow \square + {}^{4}_{2}He$$

8 Technetium-99m emits gamma radiation. Explain why the element does not change. **[2 marks]**

REVIEW LATER

# Revise it

Use this box after the lesson and before tests.

## SUMMARY

- Symbols: alpha $^{4}_{2}\mathrm{He}$, beta $^{0}_{-1}\mathrm{e}$, gamma $\gamma$, neutron $^{1}_{0}\mathrm{n}$.
- Top numbers balance; bottom numbers balance.
- Alpha: mass number − 4, atomic number − 2.
- Beta: mass number unchanged, atomic number + 1.
- Gamma: no change to either number.

## COVER THE PAGE AND ANSWER

**1** What is the symbol for a beta particle?

**2** What happens to the atomic number in alpha decay?

**3** Which decay does not change the mass number or atomic number?

**4** Complete: $^{226}_{88}\mathrm{Ra} \rightarrow ? + {}^{4}_{2}\mathrm{He}$

## I CAN…

- write the symbols for alpha, beta, gamma and neutron radiation
- balance a nuclear equation
- find the new nucleus after alpha or beta decay
- explain the effect of each decay on mass and charge

REFERENCE

# Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

## The three particles

Lesson 1

| Particle | Charge | Mass | Where |
|---|---|---|---|
| **Proton** | +1 | 1 | nucleus |
| **Neutron** | 0 | 1 | nucleus |
| **Electron** | −1 | very small | energy levels |

## Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ $1 \times 10^{-10}$ m; nucleus radius < 1/10 000 of that

## Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

## Alpha scattering

Lesson 3

| Observation | Conclusion |
|---|---|
| **most straight through** | mostly empty space |
| **some deflected** | positive centre |
| **a very few bounced back** | mass in a tiny centre |

## Models and energy levels

Lesson 4

- 1897 electron → 1904 plum pudding → 1911 nuclear → 1913 Bohr → 1932 neutron (Chadwick)
- **Absorb** EM radiation → higher level, further out.
- **Emit** EM radiation → lower level, closer in.

## Nuclear radiation

Lesson 5

- Unstable nuclei decay at **random** to become more stable.
- **Activity**: decays per second, in becquerel (Bq). **Count rate**: decays recorded per second by a detector.
- α: 2p + 2n; β: fast electron, a neutron → proton; γ: EM radiation; n: a neutron

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## Nuclear equations

Lesson 6

| Emitted | Symbol | Mass no. | Atomic no. |
|---|---|---|---|
| alpha | $^{4}_{2}\mathrm{He}$ | − 4 | − 2 |
| beta | $^{0}_{-1}\mathrm{e}$ | no change | + 1 |
| gamma | $\gamma$ | no change | no change |
| neutron | $^{1}_{0}\mathrm{n}$ | − 1 | no change |

Top numbers balance; bottom numbers balance.

$$^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

$$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$$

REFERENCE

# Key words from this lesson.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

**Daughter nucleus**: The new nucleus formed when a nucleus decays.

**Nuclear equation**: An equation showing a decay, in which the mass numbers and atomic numbers balance.

CHECK YOURSELF

# Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

### DO NOW

1. 2 protons and 2 neutrons (a helium nucleus).
2. A neutron turns into a proton, and a fast electron is emitted.
3. Becquerel (Bq).

### COVER THE PAGE

1. $^{0}_{-1}\mathrm{e}$
2. It decreases by 2.
3. Gamma emission.
4. $^{222}_{86}\mathrm{Rn}$